

Case study for GBAS Operation in Equatorial Region on Nominal day at Daytime

Hye Yeon Chang and Jiyun Lee

*KAIST / Aerospace Engineering department, Daejeon, South Korea
hyeyeonchang@kaist.ac.kr*

I. Introduction

Ground-based augmentation system (GBAS) is the system that support airplanes' precision approach and automatic landing with data obtained from GBAS ground stations. The differential ionospheric delay accounts for a large part of the error factors. Assuming that the user experiences the same ionosphere with GBAS ground stations when they are close to each other, GBAS provides the correction information obtained by ground stations to user. Therefore, by the provided correction information, user can calculate more accurate position. However, there could be residual errors of ionospheric delays. To cover the residual errors, the protection level (PL) is calculated. The protection level is the boundary that covers the error between user's estimated position and real position. The vertical protection level (VPL) is the vertical component of the protection level. The standard deviation of vertical ionospheric gradient (σ_{vig}) accounts for a large part of calculating VPL. Therefore, for the user's safety, it is necessary to provide accurate VPL with accurate σ_{vig} .

Currently, several countries located in mid-latitude region provide nominal σ_{vig} ; 4 mm/km for Conterminous United States (CONUS) [1], 2.07 mm/km for Germany [2], and 6.02 mm/km for Japan [3]. However, the geomagnetic equatorial regions have properties that ionospheric events are more active than the mid-latitude region. And also the ionospheric events are different with daytime and nighttime. Therefore, there would be limitation on providing σ_{vig} used in the mid-latitude regions to the geomagnetic equatorial regions. Also, it is necessary to calculate σ_{vig} values by distinguishing the time zone.

In this paper, case study for obtaining σ_{vig} of nominal days' daytime at Brazil region was done. Section II explains the method of date selection and ionospheric gradient calculation. Section III presents the results and discussion from a case study.

II. Simulated dates and Ionospheric gradient calculation method

A. Date Selection

The dates for the research were selected within the dates selected by INPE and Boston College research teams. The research teams selected approximately 100 dates that are considered to be a threat to GBAS operation in Brazil region, based on the geomagnetic index and scintillation index within 2011~2014 [4]. The ionospheric gradients which could be considered as ionospheric anomalies, occurred mostly during the nighttime, and during the daytime there were no ionospheric anomalies. To provide conservative σ_{vig} for nominal days' daytime, the dates simulated in this research were selected within the 100 dates which the daytime have nominal condition but nighttime are likely to be ionospheric anomalies.

One of the characteristics of the geomagnetic equatorial region is the high incidence of scintillation events. Scintillation event is a phenomenon caused by ionosphere irregularities. This event could cause refraction and diffraction of satellites' signal, so the event should be considered for the ionospheric error modeling. If the scintillation index, S4 is larger than 0.2, it can be determined that scintillation event has occurred. In this paper, the concept of scintillation occurrence rates (SOR) was used to grasp the degree of scintillation event. The scintillation occurrence rate is a percentage

of the number of data that the scintillating event have occurred($S4 \geq 0.2$), with the total number of data [5].

To consider variety degree of scintillation event during the daytime, the date has low SOR, medium SOR and high SOR were selected for calculating σ_{vig} . Table 1 shows the properties of the each selected dates.

Table 1 Properties of the selected dates

Date (year-day of year)	Category	SOR (%)
April 19 th 2012 (2012-110)	Non-scintillating day	0.1298
March 7 th 2014 (2014-66)	Scintillating day	0.3784
March 7 th 2012 (2012-67)	High daytime SOR day	2.0674

B. Ionospheric gradient calculation – Tim-step method

There are two methods for calculating ionospheric gradient; 1) Station pair method, 2) Time-step method. The station pair method is the method that calculates ionospheric gradient with the differential delays between two stations with same satellite. The time-step method configures a single satellite and a single receiver as a pair. The differential delays are calculated by comparing ionospheric delay from different time epochs. Usually many countries use the station pair method to obtain ionospheric gradient. However, for the Brazil region, it has insufficient number of stations which are located in short distance, so there is a limitation on Brazil region to use station pair method for calculating ionospheric gradient. Therefore, using time-step method to calculate ionospheric gradient on Brazil region is needed. By this method, ionospheric gradient for shorter IPP (Ionospheric pierce point) distance than the actual distance could be obtained.

The time-step method has problem that it includes not only spatial gradient but also temporal gradient. To eliminate this temporal effects, we compare the differential ionospheric delays obtained by different time window values; 1) 0.5 minute, 2) 1 minute, and 3) 1.5 minutes. According to Figure 1, there is a linear pattern in mean differential delays as time window value increase. As the time window value increase by 30 seconds, the mean vertical differential delay (dI) increase by 0.005m linearly. Therefore, temporal gradient about 10 mm/min could be included. In this paper, the standard deviation of vertical ionospheric gradient was calculated with removing some temporal effects.

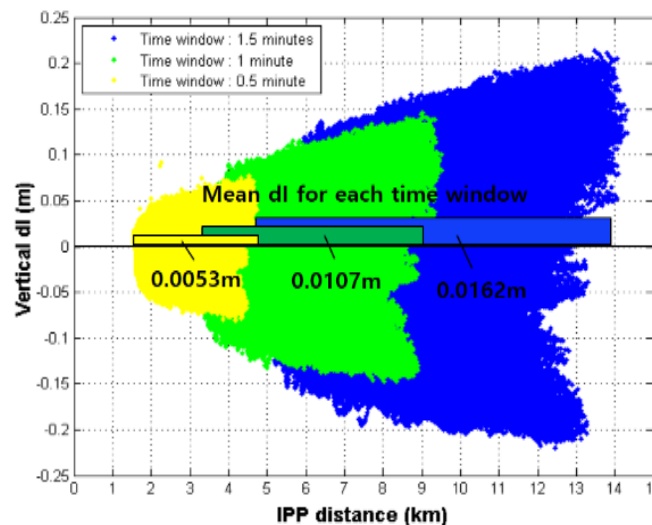


Figure 1 Mean differential ionospheric delay for each time window values

III. Results and Discussion

For obtaining protection level, zero-mean normally distributed error model is used. Therefore, CDF-bounding (Cumulative distribution function bounding) method was used to cover actual distribution by inflating 1 sigma.

There were insufficient number of data in IPP distance 6~8km. And also, the smaller the IPP distance, the larger the effect of temporal gradient, as the IPP distance small the temporal variation effect could be large. Therefore, in this paper, IPP distance between 4~6km is considered for obtaining over-bounded σ_{vig} . The results of the over-bounded standard deviation of vertical ionospheric gradient for each days are in Figure 2. Table 2 shows the summary of the results.

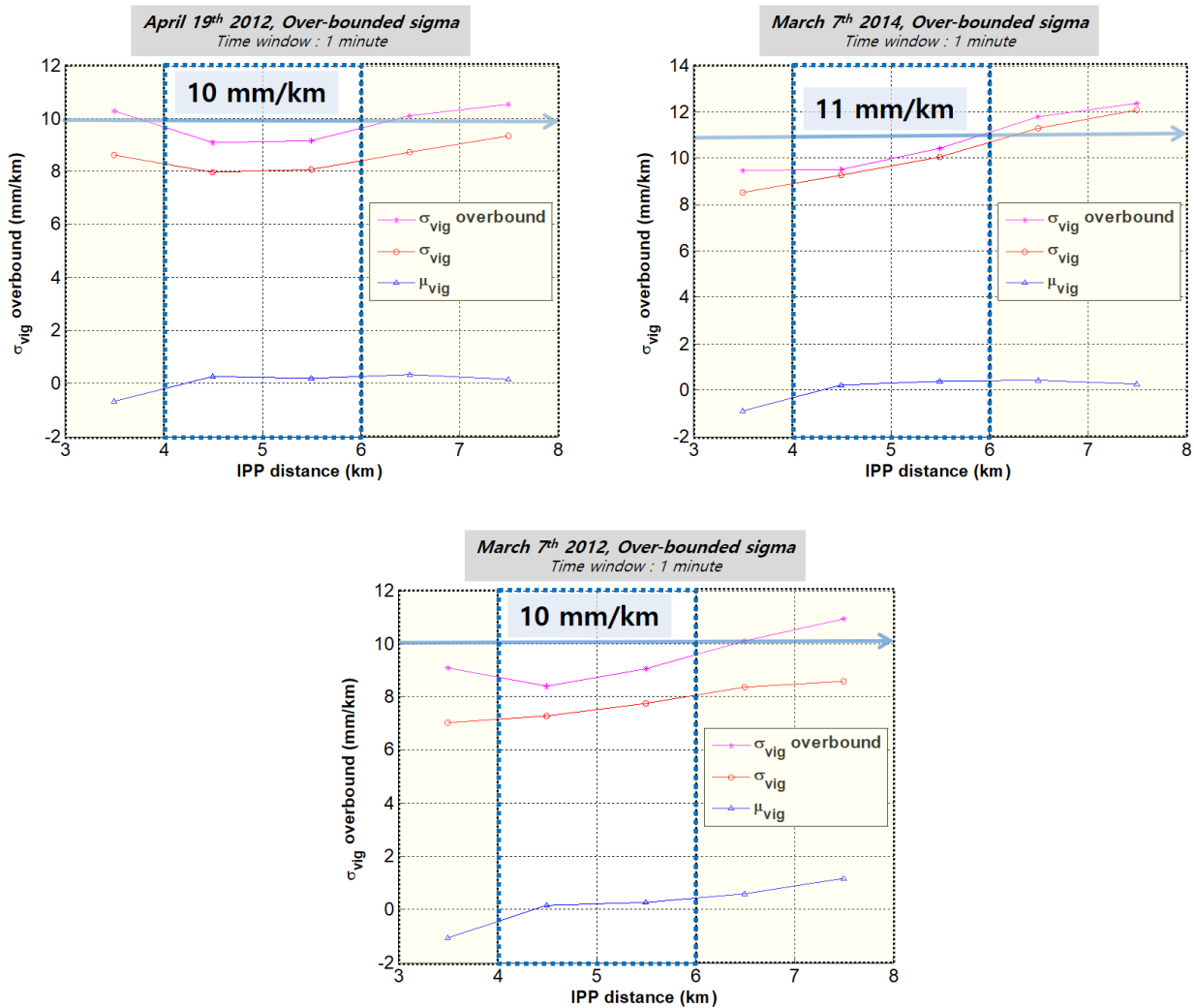


Figure 2 Over-bounded sigma (σ_{vig}) graphs for each days
(First left: April 19th 2012, First line right: March 7th 2014, Second line: March 7th 2012)

Table 2 Summary of the results

Date (year-day of year)	Category	Over-bounded σ_{vig}
April 19 th 2012 (2012-110)	Non-scintillating day	10 mm/km
March 7 th 2014 (2014-66)	Scintillating day	11 mm/km
March 7 th 2012 (2012-67)	High daytime SOR day	10 mm/km

According to Figure 2, regardless of the high or low scintillation occurrence rates, the standard deviation of vertical ionospheric gradient at nominal days' daytime was found to be about 10 mm/km. The results support the conclusion that for the daytime, the change of the standard deviation of vertical ionospheric gradient according to the degree of scintillation event was not large. Future studies will analyze additional days and calculate the standard deviation of vertical ionospheric gradient for nominal days' nighttime.

Acknowledgement

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